

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING
2019, ICRTAC 2019

Hybrid Segmentation and Feature Extraction Approach to Detect Tumour Based on Fuzzy Rough-in Mammogram Images

M Punitha* and K Perumal

Madurai Kamaraj University, TamilNadu, India.

Abstract

Tumor classification plays a significant area of research in mammogram images. In this paper, we introduce the tumor classification method in mammogram images by using Fuzzy rough set theory (FRST) and it offers an accurate approach of texture and feature extraction. The core purpose of deploying FRST is feature extraction which is achieved by using a quick reduct algorithm which helps to identify the tumor without loss of pixels in a short period. Fuzzy rough instance selection (FRIS) is applied to remove the noise from the mammogram image and finally the combination of fuzzy-rough nearest neighbor (FRNN) method is used in segmentation. The results obtained using the proposed methods are compared in various performance measures such as accuracy, sensitivity and specificity are calculated accurately.

© 2019 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019.

Keywords: FRST, Quick Reduct, FRIS, FRNN, Image Segmentation, and Feature Extraction

1. Introduction

Image segmentation is generally a very challenging problem in today's medical image processing [1] some of the computer intelligence paradigms that help segmentation process in image processing are rough set, fuzzy and fuzzy rough set [2]. Rough Set Theory (RST) introduced by Pawlak is a mathematical tool to analyze ambiguity systems

*Corresponding author. Tel.: +91-9894636960;

E-mail address: punitha_2883@rediff.com

[3][4][5] This theory was rapidly exploited and further developed for image processing and image understanding problems. The RST has quick reduction algorithms which are used to find the tumor in a short time. Many algorithms used in the past decade are based on the nearest neighbor, fuzzy theory, and rough sets. But this new approach gives a better result compared to other proposed methods. In this FRST performs the accurate research in mammogram tumor detections. The Combination of FRIS and FRNN helps to detect tumors and make segmentation process easily. The main aim of the fuzzy rough instance selection is to remove the noise from the mammogram image. This algorithm gives a positive region as a measurement tool for selecting instances.

2. Literature Survey

Senthilkumaran et.al [6] proposed the RST for Medical Image Segmentation and Granular computing. Aboul Ella Hassanien et.al [7], focused on Computational Rough Sets and Medical Imaging Segmentation and Classification, Erkan Cheng et.al [8] worked in various segmentation techniques by using Fuzzy logic. Kumar Singh et.al [9] worked as new novel approaches of segmentation concepts. El idris el kaitouri et.al [10], the process was based on fuzzy logic with image processing methods in segmentation and removes the pectoral muscles. El idris el kaitouri et.al [11] worked automatic tumour detection methods. El moufid et.al [12] focused on the region of interest in an image. Germon et.al [13] focused the stochastic relaxation. Gumei et.al [14] mainly concentrated on breast segmentation using a k-means algorithm. Chen et.al [15] focused breast region with pectoral muscle removal in a mammogram image.

3. Proposed Work

In this research work, we introduce new techniques which are called FRST (Fuzzy Rough Segmentation technique) to the mammogram images. The feature extraction is done by using the FRIS method (Fuzzy Rough instance selection). The main goal of this FRIS is to remove the noise from the mammogram image. After noise removal, the tumour is segmented by using FRNN (Fuzzy Rough nearest neighbour) algorithms. This helps to find the tumour and segmentation is done in a short period. The resultant image is classified by the evaluation technique.

4. Proposed Methodology

The FRST Segmentation techniques should be processed via various segmentation algorithms and finally classify the resultant image.

Phase 1: The pre-processing steps (i.e. fuzzy rough image pre-processing and using filters)

Phase 2: Feature extraction by using the FRIS method with a quickly reduce algorithm.

Phase 3: Extracted image should be segmented by using the FRNN method.

Phase 4: Classification has been done and evaluates the query image.

The main goal of this proposed method is to detect the tumor in a short time with cleared noise.

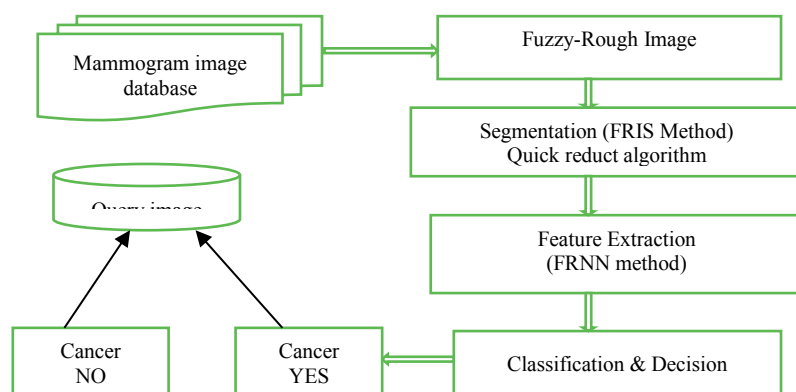


Fig 1: Flow Diagram of FRNN Method

4.1 Fuzzy rough pre-processing

Every image processing technique should have a pre-processing method as it helps to reduce the complexity of an image. In FRST, the greyscale conversion should be done in image boundaries and computed with lower and upper approximations. Here, the outer and inner approximations of an image region are done in terms of granules. Let's consider the full mammogram image as universe U and that images have comprised of a collection of pixels. At that point, we partition U into a gathering of non-covering windows of size $M * N$ each window can be considered as a granule G , this image has been segmented easily.

4.2 Segmentation with FRIS

Innumerable segmentation techniques have been done in image processing, but in this FRST image segmentation, a new approach called the FRIS (Fuzzy Rough instance selection) method which aims to remove noise and segment the mammogram image within a stipulated period. Here FRIS is applied and quickly reduce algorithm is used to segment the images.

Algorithm 1. Fuzzy Rough Instance Selection

```

Step 1: Input: A image  $A = (U, A \cup \{d\})$ ;
Step 2: A granularity parameter  $\alpha$ ;
Step 3: A threshold value  $T$ 
Step 4: Output: A Selected image  $Y \leftarrow U$ ;
Step 5: For each  $c \in U$  do
Step 6: If  $(\text{pos } A_{\alpha}(x) < T)$  then  $Y \leftarrow Y - x$ 

```

The calculation utilizes the accompanying incongruity connection:

$$R \propto a(x, y) = \max(0, 1 - \alpha |a(x) - a(y)| / I(a))$$

Where $\propto$ is a degree of granularity and $I(a)$ is the scope of the sub-image In request to choose objects the calculation offers direct multifaceted nature as for the number of items in the picture database, while calculation of the positive locale needs $O(|A| \cdot |U|)$

According to this algorithm, the instantly selected image has less threshold value. Because the predicted image always having the low pixel range the fuzzy rough instance selection method is applied by deploying the quick reduct algorithm to get the image in minimum time. For the timing purpose, quick reduct algorithm is included.

Algorithm 2: quick reduct

```

Step 1: input: A image  $A = (U, A, U \cup \{D\})$ 
Step 2: output: A supper reducts Image SR
Step 3:  $SR \leftarrow \{\}$ 
Step 4: repeat
Step 5:  $T \leftarrow SR$ 
Step 6: for each  $x \in (A - SR)$  do
Step 7: if  $Y_{SR \cup \{x\}}(x) > Y_T$  then  $T \leftarrow SR \cup \{x\}$ ;
Step 8:  $SR \leftarrow T$ ;
Step 9: end
Step 10: until  $Y_{SR} == Y_A$ ;

```

Based on FRST, the Fuzzy Quick Reduct algorithm helps to minimize the execution time and we can easily segment and get the noiseless image within a short time.

4.3 Feature Selection with FRNN

Feature selection is a process of finding features from the input image. In FRST the process of feature determination refers to finding an image reduct which is nothing but the collection of same patterns or clusters from the original image. Here Fuzzy-rough nearest neighbor (FRNN) algorithm is deployed. It is similar to k-nearest neighbors based on FRST and it predicts the new sub-image in the most similar or nearest k patterns in training data. K is randomly selected from the reduct image and calculates the tumor portion with a neighboring pixel of the reduct image. As FRNN concentrates mainly on the nearest neighboring pixel, the tumor portion can be extracted accurately.

Algorithm 3. Fuzzy Rough Nearest Neighbour

Step 1: Input: A image $A = (U, A \cup \{d\})$ as a trained image; Y is new image
 Step 2: output: predicted image $n \leftarrow \text{NN}(Y, K)$; / * NN is the k- nearest neighbour
 Step 3: $T \leftarrow 0$; p.image $\leftarrow 0$;
 Step 4: for each $c \in d$ do
 Step 5: if $((r \downarrow c(y) + (R \uparrow C)(Y))/2 \geq T$ then class $\leftarrow c$;
 Step 6: $T \leftarrow ((R \downarrow C)(Y)) + (R \uparrow C)(Y)/2$;

According to the algorithm, the nearest neighbor of each pixel is calculated and the best feature is chosen and it reduces the training time and the evaluation time to detect the features from the mammogram image.

5. Classification and Decision

Classification done for the mammogram image identifies whether it contains a tumor or not. Upon identifying

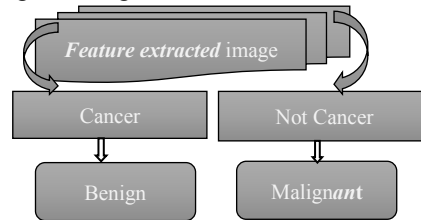


Fig 2: Flow Diagram of the Classification process

it checks for benign or malignant. Totally 60 feature extraction images and 30 images are taken as training set remaining 30 images are taken for the testing set. Then we have to predict the image which is the images affected by a tumour or not. If the image contains a tumour it is checked for being benign or malignant. The prediction is done by the below-mentioned formula.

$$\text{Predictive value} = \frac{\text{Original Value} - \text{Predictive Value}}{\text{Predictive value}} \quad (1)$$

The result is diagnosed by PPV (positive pixel value) and NPV (negative pixel value). If the resulting image gets the positive value it is called cancer image or else it gets negative value no cancer image likewise predictive values are calculated. From this cancer image, we should identify the benign and malignant by applying the below formula to identify.

$$\text{Sigmoid function} = g = 1 / (1 + \exp(-z)); \text{ Then} \quad (2)$$

Prob = sigmoid $[1 * t * m] * \theta$ g- Grad, Prob-probability, T- Thickness of clump
 M-Marginal adhesion.

Our count of the probability of whether a tumour is dangerous or amiable depends on 2 highlights, clump thickness, marginal adhesion. With suitable qualities substituted in the equation, the probability if a tumour is harmful or amiable is determined.

6. Result and Discussion

All the images are pre-processed and the FRST segmentation techniques are applied which help to reduce the noise level for every mammogram image and finds the tumor utilizing segmentation and extraction with the help of FRIS and FRNN. It gives the high quality in different images then classification has been done by prediction values. The proposed method gives the better result and there is no other research has been done.

6.1 Database

The MiniMIAS [16] database containing left and right breast images for a total of 60 patients is used in this study. All images containing hypothesized masses and a selection of normal types are considered. All images of 1024 X 1024 pixels and at 8-bit greyscale level. In the analysis of results within the study, and we use the following methods.

6.2 Tools

There are so many numbers of tools available that are compatible with image processing for performing segmentation. Here we use R tool for efficient implementation

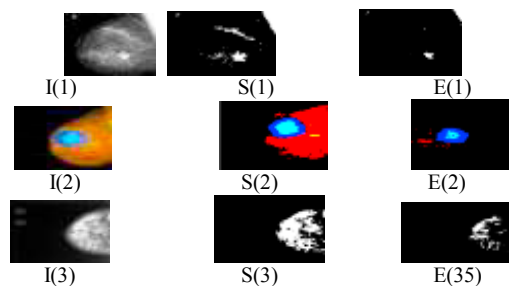


Fig 3: Mammogram Images Segmentation and Extraction Result

Figure 3: I (1) to I (3) in the 1st column are Input of mammogram image, which should contain the tumor. S (1) to S (3) in the 2nd column is the results of FRIS segmentation. E (1) to E (3) in the 3rd column is the tumor extracted from the segmented mammogram images and the experimental result can be evaluated in quality measures such as sensitivity, specificity, accuracy are computed and compared with some other existing methods.

Sensitivity: the ratio of the number of detected pixels as a tumor to the whole number of tumor pixels in the image.

- $$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (3)$$

- Specificity: the ratio of the number of detected pixels as clean to the whole number of clean pixels in the image.

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (4)$$

- Accuracy: To estimate the exactness of a test, we should calculate the proportion of true positive and true negative in all evaluated images, and it can be defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

They are computable according to the following formulas and the presence of the objective abnormality.

- True Negative (TN): the tested result is negative values.
- False Positive (FP): the tested result is a positive abnormality.
- False Negative (FN): the tested result is undesirable in the presence of the clinical abnormality
- FP= False Positive value should be calculated
- FN= False Negative value should be Calculated

Table 1: Comparison of various segmentation Technique

<i>Comparison of classification performance measures based on various segmentation techniques.</i>				
<i>Images</i>	<i>Segmentation Methods</i>	<i>Specificity</i>	<i>Sensitivity</i>	<i>Accuracy</i>
<i>Image(1)</i>	<i>FRST segmentation</i>	89.23%	91.23%	97.45%
<i>Image(2)</i>		76.54%	67.45%	92.58%
<i>Image(3)</i>		72.45%	67.25%	94.23%
<i>Image(4)</i>		78.45%	72.53%	99.02%
<i>Image(5)</i>		89.45%	93.45%	97.45%
<i>Image(1)</i>	<i>Rough segmentation</i>	79.45%	80.25%	82.45%
<i>Image(2)</i>		69.45%	72.47%	81.20%
<i>Image(3)</i>		69.56%	72.26%	78.15%
<i>Image(4)</i>		82.12%	83.45%	90.21%
<i>Image(5)</i>		78.45%	58.56%	60.23%
<i>Image(1)</i>	<i>Fuzzy segmentation</i>	89.12%	89.89%	90.12%
<i>Image(2)</i>		78.63%	88.45%	91.21%
<i>Image(3)</i>		60.75%	72.45%	80.12%
<i>Image(4)</i>		82.35%	83.24%	75.45%
<i>Image(5)</i>		80.45%	85.68%	89.12%

As for this table 1 shows the various comparison results of segmentation techniques and the proposed methods should give better results compared than the other two methods. Especially the accuracy should be nearer the 99.02% in the proposed method and the specificity; sensitivity also had good performance measures. On seeing the overall performance, the new FRST segmentation proves better than the other methods compared above.

7. Conclusion

The main goal of this proposed method is to find the tumour without loss of pixel from the image in a short time. In this method, a hybrid combination of the fuzzy rough instance selection (FRIS) and Fuzzy rough nearest neighbour (FRNN) is used for segmentation. The quick reduct algorithm reduces the computation time. The tumour classification is done by the prediction probability of PPV (positive pixel value) and NPV (negative pixel value). The probability calculation is made by regardless of whether a tumour is threatening or amicable and that depends on 2 highlights, clump thickness, and marginal adhesion. With right qualities substituted in the equation probability of finding whether a tumour is harmful or favourable is achieved. On the off chance that the probability is over half, the tumour is threatening (destructive), and if the likelihood is under half, it is benevolent (non-dangerous) effectively. Therefore the result shows the improved mutation of the high-performance computing technology and obtains 99.02% accuracy. Besides, this combination of the proposed work is much useful in medical image diagnosis and analysis.

References

- [1] Bouchet A, Pastore J, and Ballarin V, "Segmentation of Medical Images using Fuzzy Mathematical Morphology", *JCS and T*, Vol.7, No.3, October 2007, pp.256-262.
- [2] J. Bezdek, "What is computational intelligence?" in *Computational Intelligence: Imitating Life*, J. Zurada, R. Marks, and C. Robinson, Eds. Piscataway, NJ: IEEE Press, 1994, pp. 1–12.
- [3] Pawlak. Z and Slowinski. R, "Rough Set approach to multiattribute decision analysis", *Invited Review, Eur.Journal of Oper. Res.*, Vol.72, 1994, pp. 443-459.
- [4] Komorowski.J, Pawlak.Z, Polkowski.L,Skowron.A, "Rough Sets: A tutorial",In *Pal.S.K,Skowron.An (Eds) Rough Fuzzy Hybridization: A new trend in Decision making*, Springer, Singapore, 1999, pp.3-98.
- [5] P.Chen, G.Wang, Y.Yang and J.Zhou, "Facial Expression Recognition based on Rough set theory and SVM", *LectureNotes in Computer Science, Springer Berlin/ Heidelberg, Rough sets and knowledge Technology*, Vol. 4062, 2006, pp. 772-777.

- [6] N. Senthilkumaran and R. Rajesh, "A Study on Rough Set Theory for Medical Image Segmentation", *International Journal of Recent Trends in Engineering*, Vol 2, No. 2, November 2009
- [7] Aboul Ella Hassanien, Ajith Abraham, *Senior Member, IEEE*, James F. Peters, *Senior Member, IEEE*, Gerald Schaefer, *Member, IEEE*, Christopher Henry, *Member, IEEE*, *Rough Sets and Near Sets in Medical Imaging: A Review*, 2009.
- [8] Erlang Cheng et al, "Mammographic image classification using histogram intersection", *Biomedical Imaging: From Nano. to Macro*, IEEE, pp.197 – 200, May 2010.
- [9] Kumar Singh, A., and Gupta, B., *A Novel Approach for Breast Cancer Detection and Segmentation in a Mammogram* Elsevier page No 676–682, 2015. 1877-0509 © 2015 The Authors. Published by Elsevier B.V. This is an open-access article under the CC BY-NC-ND license
- [10] El Idrissi el Kaitouni S, Abbad A, Tairi H (2017). Tumor extraction and elimination of pectoral muscle based on hidden Markov and region growing: applied based MIAS. In the International Conference on Advanced Technologies for Signal and Image Processing (ATSIP). IEEE Conference.
- [11] El Idrissi el Kaitouni S, Abbad A, Tairi H (2017) Automatic detection of the tumor on mammogram images based on hidden Markov and active contour with quasi-automatic initialization. *International Journal of Medical Engineering and Informatics* 9(4):316–331, volume 9, Issue 4.
- [12] Elmoufidi A, El Fahssi K, Jai-Andaloussi S, Madrane N, Sekkaki A (2014). Detection of regions of interest's in mammograms by using local binary patterns, dynamic k-means algorithm, and gray level co-occurrence matrix. In *Next-Generation Networks and services (NGNS)*, 2014 Fifth International Conference on (pp. 118–123). IEEE
- [13] Geman S, Geman D (1984) Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images. *IEEE Trans Pattern Anal Mach Intell* 6:721–741
- [14] Gumaei A, El-Zaar A, Hussien M, Berbar M (2012) Breast segmentation using a k-means algorithm with a mixture of gamma distributions. In *Broadband Networks and Fast Internet (RELABIRA)*, 2012 Symposium on (pp. 97–102). IEEE
- [15] Chen Z, Zwiggelaar R (2010) Segmentation of the Breast Region with Pectoral Muscle Removal in Mammograms. *Medical Image Understanding and Analysis (MIUA) 2010*. The University of Warwick, Coventry, pp 71–76
- [16] The mini-MIAS Database of Mammogram, <http://peipa.essex.ac.uk/info/mias.html>